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Weak polyelectrolyte complexation driven by associative charging

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Weak polyelectrolytes are relevant for a wide range of fields; in particular, they have been investigated as “smart” materials for chemical separations and drug delivery. The charges on weak polyelectrolytes are dynamic, causing polymer chains to adopt different equilibrium conformations even with relatively small changes to the surrounding environment. Currently, there exists no comprehensive picture of this behavior, particularly where polymer–polymer interactions have the potential to affect charging properties significantly. In this study, we elucidate the novel interplay between weak polyelectrolyte charging and complexation behavior through coupled molecular dynamics and Monte Carlo simulations. Specifically, we investigate a model of two equal-length and oppositely charging polymer chains in an implicit salt solution represented through Debye–Hückel interactions. The charging tendency of each chain, along with the salt concentration, is varied to determine the existence and extent of cooperativity in charging and complexation. Strong cooperation in the charging of these chains is observed at large Debye lengths, corresponding to low salt concentrations, while at lower Debye lengths (higher salt concentrations), the chains behave in apparent isolation. When the electrostatic coupling is long-ranged, we find that a highly charged chain strongly promotes the charging of its partner chain, even if the environment is unfavorable for an isolated version of that partner chain. Evidence of this phenomenon is supported by a drop in the potential energy of the system, which does not occur at the lower Debye lengths where both potential energies and charge fractions converge for all partner chain charging tendencies. The discovery of this cooperation will be helpful in developing “smart” drug delivery mechanisms by allowing for better predictions for the dissociation point of delivery complexes. *Published by AIP Publishing.* <https://doi.org/10.1063/1.5017941>

INTRODUCTION

Polyelectrolytes are macromolecules in which a substantial portion of the constituent units includes groups which are ionic or ionizable.¹ These molecules derive important functionality and assembly properties from both their electrostatic and polymeric character. Abundant in nature,² polyelectrolyte-driven assembly has recently seen use in a variety of synthetic applications, including the creation of anti-fouling coatings to protect underwater surfaces from destruction by sea organisms,³ and in the creation of selective membranes.^{4–7} Polyelectrolytes have also been known to exhibit unique complexation characteristics,^{8,9} permitting the formation of strongly adhering layer-by-layer assemblies¹⁰ as well as polymer-rich liquid phases known as coacervates that have attracted intense recent interest^{11,12} for their capabilities in food processing and drug delivery, among other applications.^{13–15}

Two regimes, strong and weak, are often used to categorize polyelectrolyte materials. Strong polyelectrolytes are polymers whose monomers consist of strong acids or bases and are thus fully charged at all commonly encountered pH values. By contrast, weak polyelectrolytes contain weakly

acidic or basic monomers, and thus the degree of charging is sensitive to the value of pH.¹⁶ While the charges of strong polyelectrolytes are fixed to specific sites on the polymer, weak polyelectrolytes may anneal—with charges moving between ionizable groups—in response to the local energetic environment.¹⁷ This property imbues weak polyelectrolytes with interesting charge-driven behavior, particularly when charge repulsions compete with adhesion to determine molecular conformations. Near the pK_a (or pK_b), the fraction of charged sites on a single molecule is distributed to maximize adhesive contacts, leading the polymer to adopt compact conformation. These competing interactions result in a sharp, first order coil to globule transition for large polymers.^{18,19} Similar effects of pH on chain conformation have also been used to create smart, pH-sensitive membranes.²⁰

Polyelectrolyte complexes have also been utilized for drug delivery purposes. For example, polyelectrolyte capsules fabricated by layer-by-layer deposition of oppositely charged polyelectrolytes around a charged core template may be loaded with drug molecules after removal of the core.²¹ The properties of such capsules depend on the polyelectrolyte layers used to form them, and hence weak polyelectrolyte complexes with stimuli-responsive charging behavior can be utilized to tune the stability of a capsule at a target pH. Furthermore, to enhance biocompatibility, biologically derived molecules may also be

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incorporated into these capsules.²² One successful application was reported by Shen and co-workers,²³ who utilized doxorubicin loaded bovine serum albumin (BSA)-gel capsules for the inhibition of the pulmonary melanoma growth. These capsules displayed excellent pH-controlled loading and release properties. Additionally, Anandhakumar *et al.*²⁴ designed a polyelectrolyte thin multilayer biocompatible film with epithelial cells using a layer by layer assembly of polyallylamine hydrochloride (PAH) and polymethacrylic acid (PMA). The film was loaded with bovine serum albumin (BSA) and ciprofloxacin hydrochloride (CH), and the amount of release was controlled by changing the environmental conditions such as pH and ionic strength.

Though weak polyelectrolytes are often used in applications,²⁵ a comprehensive conceptual picture uniting their charging properties to the influence of other environmental factors has yet to emerge. This absence hampers our ability to engineer and optimize the behavior of a polyelectrolyte system for a given task. Molecular simulations present a fantastic tool by which we understand the physical mechanisms of charging in weak polyelectrolytes. There exists extensive theoretical and simulation literature on the topic strong polyelectrolytes, and their behavior in dilute and semi-dilute conditions is well understood, both with and without the addition of salt.^{26,27} The behavior of complexing oppositely charged polyelectrolytes is less well understood and has thus far focused on highly idealized models in solutions with monovalent salts, thus permitting comparison to continuum theoretical results.^{8,28,29} This situation (layer-by-layer assembly or complex coacervation, depending on the strength of electrostatic coupling which drives phase separation) has seen a recent explosion of interest driven by the novel phase separation and response properties that could be exploited, e.g., underwater adhesives^{30–32} and drug delivery capsules.^{33–37} For coacervating phases, the use of low-charge-density or weak polyelectrolytes is essential, as strong polyelectrolytes with densely arrayed charges tend to form into solid aggregates rather than fluid phases.^{38–40} To properly characterize and control these phases, it is imperative that one understands the charging behavior of weak polyelectrolyte systems, particularly in dense and confined conditions. Within the current work, we aim to understand the interplay between weak polyelectrolyte charging and complexation behavior. While some prior work has been performed which characterizes the charging tendency of single weak polyelectrolyte chains in solution,^{41,42} little has been done to understand how these polyelectrolytes will be affected by their local environment, in particular due to the presence of oppositely charged electrolytes.

It is clear that the structure of a solvated weak polyelectrolyte in solvent is dictated by a tug-of-war between three primary contributions: (1) the solvation properties of the polymer backbone, (2) the strength of electrostatic forces in the system, and (3) the acidity or basicity of the polymer chain relative to ambient pH. The first two conspire in the strong polyelectrolyte limit to give a variety of known morphologies including the coil, globule, and pearl necklace configurations,^{27,43} which are selected based on the spacing of charges and the presence of salt. In the latter case, extensive work^{44–46} has demonstrated that strong polyelectrolytes

undergo collapse to a globule state at high salt concentrations, and potentially a re-expansion at higher concentrations, under conditions which depend on the valence of salt counterions.⁴⁴ Monovalent salts act simply to screen the overall electrostatic interactions unless concentrations are sufficient to induce strong charge-charge correlations. In the case of multivalent salts, strong coupling of multivalent counterions to the backbone of the chain acts initially to neutralize the chain, collapsing it via strong charge interactions before eventually inducing overcharging and reentrant swelling of the polymer, particularly when the salts have a valence larger than three. Further studies on counterion behavior have shown that complex, flexible chains tend to attract a larger amount of counterions into its globular conformation.⁴⁷

The role of pH in charging a weak linear polyelectrolyte has been explored in the implicit salt regime.^{18,48,49} These studies examine the conformational changes induced in a poorly solvated weak polyelectrolyte through the contributions of salt screening and pH, characterizing a transition from globular to pearl necklace-like to linear expanded conformations. As the amount of charge on the polymer increases, Coulombic repulsions begin to dominate, expanding the chain, in accordance with behaviors seen in experiment.⁵⁰ For more complex chains, such as star polymers, charges locate preferentially away from the core or other highly substituted monomers.¹⁹ It should be noted that all of these investigations focused on the properties of individual polymers. It is not clear from them what role other nearby polyelectrolytes may play in inducing charging of a weak polyelectrolyte, and the extent to which associative interactions drive (or are driven by) preferential charging of monomer sites. Utilizing a highly idealized model of weak polyelectrolytes, we investigate here the presence of cooperative charging effects in oppositely charged weak polyelectrolyte chains. The emphasis here is on determining the interplay of electrostatic screening and hydrogen activity in charging each chain and which conditions result in chain aggregation.

METHODS

Our approach uses an implicit titration of interacting weak polyelectrolytes using combined Monte Carlo and molecular dynamics (MCMD) simulations, with the role of salt accounted for by screened Debye-Hückel electrostatics. In describing our coarse-grained model, we utilize reduced units set by the length scale of monomers (σ), their mass (m), charge (q), and thermal energy ($k_B T$, where k_B is Boltzmann's constant). All simulations are performed in the NVT -pH ensemble, with the total number of monomers, volume, temperature, and pH constant in each simulation. A box size of $64.6 \sigma^3$ with periodic boundaries was utilized for all simulations.

Our model consists of two bead-spring polymer chains based on the Kremer-Grest⁵¹ and Stevens-Kremer⁵² models, with $N = 60$ monomers apiece, which are approximated by coarse-grained beads of size σ connected by finitely extensible nonlinear elastic (FENE) springs. Size and interaction parameters for charged and uncharged monomers are taken to be identical aside from the presence of a positive or negative charge. The full force field is described by Eq. (1) which has

three contributions: FENE potential (U_{FENE}),⁵³ representing a bead-spring polymer chain; shifted-truncated Lennard-Jones repulsion (U_{LJtrunc}) for steric repulsions; and Debye–Hückel approximation electrostatics (U_{coul}) for charge interactions,

$$U_{\text{total}} = \sum_{i < j} U_{\text{LJtrunc}}(r_{ij}) + U_{\text{coul}}(r_{ij}) + \sum_{(i,j) \text{ bonded}} U_{\text{FENE}}(r_{ij}), \quad (1)$$

where the indices i and j run over the beads in the system and the (i,j) in the FENE term indicates a sum over bonded pairs. Here, a shifted truncated Lennard-Jones interaction, defined as

$$U_{\text{LJtrunc}}(r) = \begin{cases} U_{\text{LJ}}(r) - U_{\text{LJ}}(r = 2^{1/6}\sigma), & \text{for } r \leq 2^{1/6}\sigma, \\ 0, & \text{elsewhere,} \end{cases} \quad (2)$$

$$U_{\text{LJ}}(r) = 4\epsilon_{\text{LJ}} \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right], \quad (3)$$

is used to prevent bead overlap with the coupling constant ϵ_{LJ} set by the thermal energy scale (Boltzmann's constant times temperature, $k_{\text{B}}T$, i.e., $\epsilon_{\text{LJ}} = k_{\text{B}}T$). The bonded potential follows the standard FENE form

$$U_{\text{FENE}}(r) = -0.5KR_0^2 \log \left[1 - \left(\frac{r}{R_0} \right)^2 \right], \quad (4)$$

where R_0 is the maximum extent of the FENE bond and K is a spring-strength constant set to 2σ and $7\epsilon_{\text{LJ}}/\sigma^2$.

The interaction between charged beads is modeled through the screened Coulomb (Debye–Hückel) potential, calculated as

$$U_{\text{coul}}(r) = k_{\text{B}}T\lambda_{\text{B}} \frac{q_i q_j}{r} e^{-\kappa r}, \quad (5)$$

where κ^{-1} is the Debye length in units of σ^{-1} . A potential cut-off, beyond which no contributions to U_{coul} were calculated, was fixed for all simulations and was set equal to $r_{\text{cut}} = 50\sigma$. As the Debye length is inversely proportional to the square root of the ionic strength, this accounts for salt-dependent screening between charged sites.⁵⁴ The remaining term λ_{B} in the equation above defines the Bjerrum length and is set to 1.0σ . This is a realistic choice, given the typical size of hydrated monovalent ions.⁵⁴

The two chains in the box are weak polyelectrolytes of opposite charge; one acquires $-e$ charge per monomer, while the other chain acquires $+e$ charge depending on the activity of solution and the local environmental conditions. The total charge on each polymer is thus a discrete value between 0 and $60e$. For clarity, we define the polymer chain that acquires $-e$ charge as the polyacid (PA) and the polymer chain that acquires $+e$ charge as polybase (PB).

We consider the environmental pH implicitly through charging chemical potentials μ_{PA} and μ_{PB} , with larger values for both corresponding to a higher fraction of monomers charged for both polyacids and polybases. Since acids (e.g., acrylic acid) exist in the protonated (uncharged) state at pH conditions below the pK_{a} of the acid and in a deprotonated ($-e$ charged) state at pH above the pK_{a} of the acid, we define charging of the polyacid as the following equilibrium reaction:



where HA represents an uncharged monomer of the coarse-grained polyacid and A^- represents a charged monomer. Therefore, the relevant mapping for μ_{PA} of a polyacid to pH is

$$\ln 10 (\text{pH} - \text{pK}_{\text{a}}) = \beta (\mu_{\text{HA}}^0 - \mu_{\text{A}^-}^0) = \beta \mu_{\text{PA}}, \quad (7)$$

where β is $1/k_{\text{B}}T$, pK_{a} is the $-\log(K_{\text{a}})$ of the monomer unit of the polyacid, and μ_{HA}^0 and $\mu_{\text{A}^-}^0$ are standard state chemical potentials of the uncharged and charged monomers, respectively. In the derivation of Eq. (7), the proton chemical potential, $\mu_{\text{H}^+}^0$, cancels, as $\text{pH} = -\log a_{\text{H}^+}$ and $K_{\text{a}} = \frac{a_{\text{H}^+} a_{\text{A}^-}}{a_{\text{HA}}}$. The charging chemical potential can be thought of as controlling the exchange of protons with the bulk solution. Similarly, charging of bases (e.g., ethylene amine) may be related to pK_{b} , or the pK_{a} of the conjugate acid, and we define the charging reaction for the polybase in terms of a dissociation reaction of the conjugate acid as the following:



where B represents an uncharged monomer of the coarse-grained polybase and BH^+ represents the charged monomer. The definition of μ_{PB} for a polybase would then be

$$\ln 10 (\text{pK}_{\text{a,conj}} - \text{pH}) = \beta (\mu_{\text{B}}^0 - \mu_{\text{BH}^+}^0) = \beta \mu_{\text{PB}}, \quad (9)$$

where $\text{pK}_{\text{a,conj}}$ is the $-\log(K_{\text{a,conj}})$ for the conjugate acid (BH^+) of the base (B) and μ_{B}^0 , $\mu_{\text{BH}^+}^0$ are standard state chemical potentials of the uncharged and charged monomers, respectively. Monte Carlo titration moves (charging or discharging of a given chain) are proposed at random, with a Metropolis acceptance rule combining the energetic difference of charging or discharging with the associated chemical potential cost. Here, the acceptance probability is given by

$$P_{\text{acc}} = \min(e^{-\beta \Delta U \pm \beta \mu}, 1), \quad (10)$$

where ΔU is the total change in potential energy during the Monte Carlo move and the generic term μ represents μ_{PA} or μ_{PB} , with equal probability of charging ($e^{+\beta \mu}$) or discharging ($e^{-\beta \mu}$) the monomers.

Furthermore, in weak polyelectrolytes, ions can dissociate and recombine freely; hence the charges are “mobile” along the chain. This property was simulated by proposing a charge swapping move between two monomers which ignores the charging cost term ($e^{\pm \beta \mu}$) in Eq. (10) and hence acceptance is determined by

$$P_{\text{acc}} = \min(e^{-\beta \Delta U}, 1). \quad (11)$$

A note to be made here is that neither the charges on the polyacid nor those on the polybase are kept fixed during the simulation and that the charging/discharging along with the charge annealing Monte Carlo moves of the polyacid and polybase is independent of each other. Furthermore, the simulations are not always charge neutral as the counterions required to satisfy charge neutrality are not fully resolved in the Debye–Hückel limit. Figure 1 diagrams the important Monte Carlo (MC) moves involved in these simulations. These moves are essential for sampling relaxation of charges along the polymer chains. However, without implementing complex arrays of moves, MC alone is unable to quickly sample the equilibrium conformation of polymer systems.⁵⁵

To improve sampling of polymer conformations, we utilize hybrid molecular dynamics simulation. We implement a

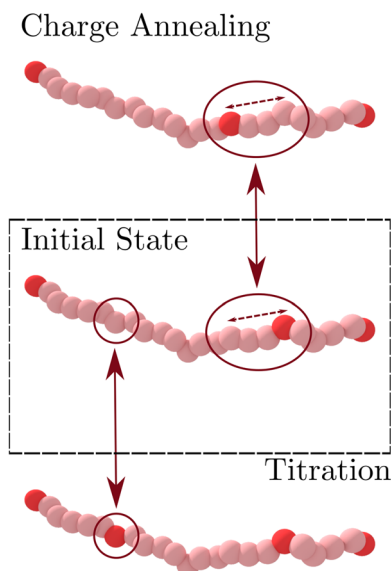


FIG. 1. Monte Carlo moves utilized in the simulations, with charged monomers colored red and uncharged monomers colored pink. The implicit titration governs the titration/charging of the polyelectrolyte, whereas charge annealing governs the movement of charges along the polyelectrolyte chain. Polymer conformations are illustrative only and not representative of states observed in simulation.

set of routines combining the open-source codes LAMMPS (MD)⁵⁶ and SAPHRON (MC)⁵⁷ and governing the exchange of information between them; the source code is available through GitHub (see the url in Ref. 57). The simulations proceed by performing a short MD run, after which MC moves are randomly attempted. For these simulations, we use short trajectories of 1000 MD steps between MC moves and attempt 20 MC moves per MD sweep (1 MD sweep = 1000 MD steps), distributed equally among annealing and titration moves. For the MD evolution in LAMMPS, the mass of all species, in reduced units, was set to 1. The time step length was $\delta t = 0.01\tau$ in reduced units, where $\tau = \sigma\sqrt{m/\epsilon_{LJ}}$. The Langevin thermostat of a damping parameter 100τ was implemented to

sample a canonical ensemble and the desired temperature was set to 1.0 in reduced units, with $\epsilon_{LJ} = k_B T$. Along with the Bjerrum length, these numbers allow us to approximately map our simulation results onto experimental systems. It should be noted that our choice of Bjerrum length approximates the value in water near room temperature if ions are assumed to be hydrated. For bare ions, it should be approximately 2.8σ if σ is taken to represent the monomer size in poly(acrylic acid) (PAA) or poly-(2-vinyl pyridine) (P2VP). We note that choosing a larger value of λ_B does not qualitatively affect these results, in fact, due to the longer range and stronger electrostatic coupling, charge repulsion and association effects should be prominent there. The total number of Monte Carlo moves performed in each simulation was $\sim 10^6$, whereas the total number of molecular dynamics steps was $\sim 5 \times 10^7$. In certain cases, outlined below, such as obtaining the average distance between the center of mass of a polyacid and polybase, longer simulations with a total of 4.5×10^6 Monte Carlo moves were performed, resulting in a total of 2.25×10^8 molecular dynamics steps. Statistics for each simulation were obtained by averaging over the last 7000 MD sweeps.

To examine the charging tendencies of this model, and qualitatively test our implementation, we initially study the titration of a single polyacid chain. Importantly, as the salt content in the solution increases, the electrostatic interactions will weaken, facilitating greater charging of the chain. We plot this in Fig. 2(a) for various κ values. For comparison, we plot the ideal-case Henderson–Hasselbalch (HH) relation between a charging potential and charge fraction. We see that the titration curves are shifted towards the right side for lower κ values, indicating suppression of charging relative to the ideal expectation. This results from an energetic penalty between neighboring like-charged groups. With an increase in κ , screening increases and monomers cease to interact strongly, after which charging behavior approaches the HH limit. Examining the variation of the “half-charge chemical potential,” $\mu_{0.5}$, with κ clarifies this behavior. At lower κ values, $\mu_{0.5}$ is significantly

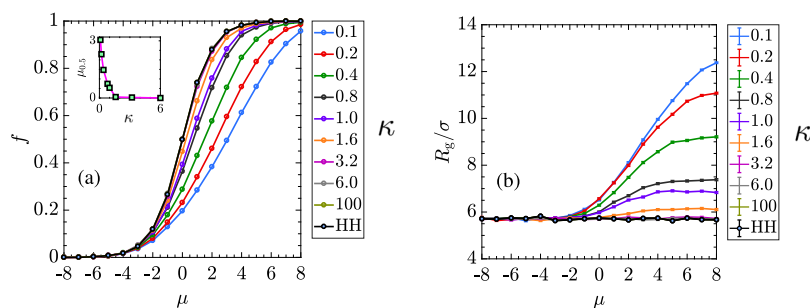


FIG. 2. (a) Titration curves for a single weak polyelectrolyte as a function of κ and μ . κ values are presented in the legend as different colored lines, whereas μ values are set corresponding to Eq. (7) and are utilized in titration Monte Carlo moves through Eq. (10). f represents the charge fraction of the isolated weak polyacid. These simulations were performed with a single $N = 60$ polyacid in the simulation box described before in the section titled Methods. HH refers to the ideal Henderson–Hasselbalch limit where the charging of monomer is not influenced by the interactions with neighboring monomers and hence at $\text{pH} = \text{pK}_a$ conditions which correspond to $\mu = 0$, 50% of the weak polyacid is charged. At lower κ values, neighboring interactions with charged monomers are not negligible and result in an increase in electrostatic potential [Eq. (5)] which suppresses the charging of the weak polyacid [Eq. (10)] and hence shifting the titration curve towards the right side (larger μ values). Inset plot depicts $\mu_{0.5}$ as a function of κ , where $\mu_{0.5}$ is the value of μ where the polyelectrolyte reaches an equilibrium charge fraction of 0.5. As $\kappa \rightarrow \infty$, the system approaches behavior described by the Henderson–Hasselbalch equation ($\mu_{0.5} \rightarrow 0$). (b) Depicts the corresponding radius of gyration of the $N = 60$ weak polyacid under the same conditions as (a). Expansion of the weak polyacid as evident from the large radius of gyration values at low κ and high μ values is due to the intra-chain repulsive interactions, as mentioned before, which causes the swelling of the polyacid chain.

shifted, ~ 3 units (or $\text{pH} - \text{pK}_a = 1.302$) away from the HH limit, while at $\kappa = 6$ we have nearly reached the asymptotic HH limit ($\mu_{0.5} \rightarrow 0$). However, charging within this limit does not lead to an extension in the polymer chain, as the long-range repulsions which inhibit charging are also essential in driving chain extension. This is evidenced by the decreased radius of gyration for a single chain at large μ observed in Fig. 2(b). The important effect to see here is that charging along an isolated chain is antagonistic, penalized by long-range electrostatics, hence it is expected that the charging of two oppositely interacting chains is cooperative. Regardless, the single-chain calculation demonstrates that the role of the charge environment is nontrivial, and salt content along with pH will both have significant bearing on the charge balance of associating weak polyelectrolytes.

RESULTS AND DISCUSSION

For the discussion below, increased screening will be represented by the increase in the value of κ and increased charging tendency by the increase in value of μ_{PA} and μ_{PB} for the polyacid and polybase, respectively, as these along with the force field described previously specify all necessary conditions for a given simulation. Figures 3 and 4 depict the average charge fraction (f_{PA}) of the polyacid in the presence

of a polybase with variation in μ_{PB} , μ_{PA} , and κ . At a charge fraction of 0, none of the monomers of the polyacid are charged, whereas at the charge fraction of 1, all the monomers of the polyacid are charged. For the case of Fig. 3, the charging behavior at all combinations of polyacid and polybase chemical potential may be inferred from the heat map for four typical values of κ . Figure 4 breaks this information down along lines of constant polyacid chemical potential near the monomer pK_a as a function of polybase strength and screening parameter, comparing to the charge curve expected for an isolated polyacid chain. Analogous plots for the polybase are given in Figs. S1 and S2 of the [supplementary material](#). Importantly, as noted in the section titled Methods, the two-chain system need not be charge neutral; a plot of the net charge is given in Fig. S3 of the [supplementary material](#).

At $\kappa = 0.1$, monomers of both the polyacid and polybase have a relatively long-range interaction, leading to significant cooperativity with monomers of each species aiding charging of each other. This may be compared to experimental conditions through the mapping described in the section titled Methods and represents screening due to a monovalent salt concentration of ~ 1.8 mM. Intriguingly, these results demonstrate that even at the conditions where it is unfavorable for a polyacid to charge in isolation (see Fig. 4), interactions with a polybase with high charging tendency leads to a

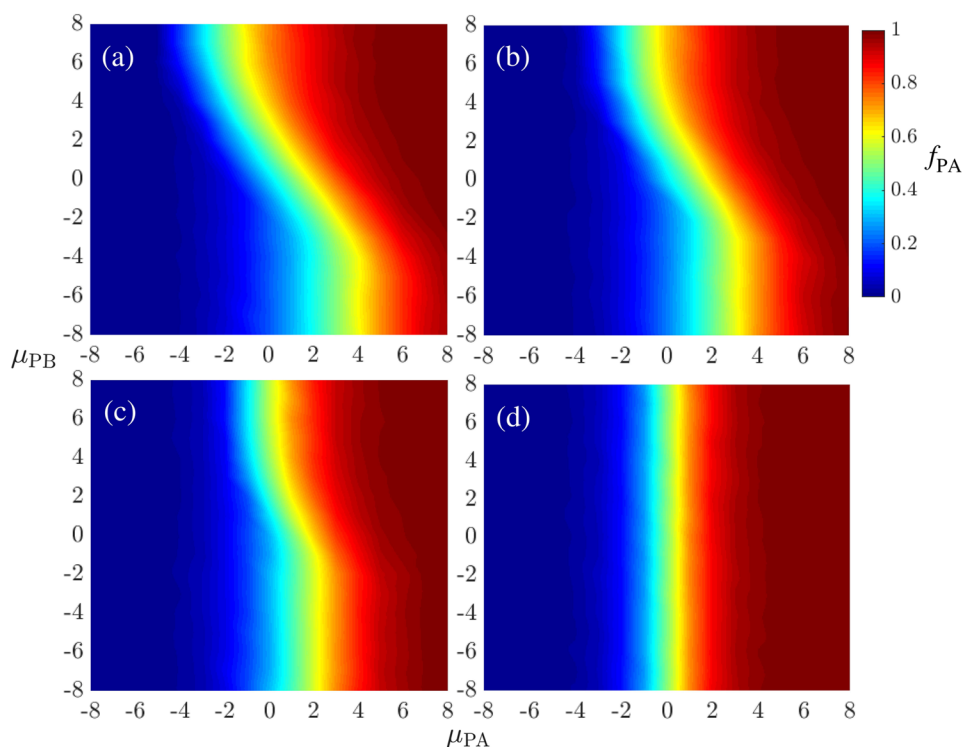


FIG. 3. Induced associative charging due to polyelectrolyte interactions. This shows the equilibrium charge fraction of the polyacid with variation of μ_{PA} (x-axis) and μ_{PB} (y-axis) at fixed κ values. The inverse of the Debye length, κ , is a representative of the electrostatic screening in the system and thereby affects the interaction between polyelectrolytes. (a) $\kappa = 0.1$, (b) $\kappa = 0.2$, (c) $\kappa = 0.4$, (d) $\kappa = 6.0$. Charging tendency is presented by μ_{PA} and μ_{PB} , which indicate a larger tendency to charge for both the polyacid (PA) and polybase (PB) at higher values, respectively. Color bar indicates the charge fraction of the polyacid (PA). It should be noted that as constructed this system is symmetric, and thus the polymer chains in the simulation can be representative of any pair of oppositely charging weak polyelectrolytes. Hence, transposing the data in (a)–(d), i.e., plotting μ_{PA} on the y-axis and μ_{PB} on the x-axis, the equilibrium charge fraction of polybase represented through colormap would look the same as the result presented in (a)–(d). To illustrate the symmetric nature, Fig. S1 of the [supplementary material](#) contains the analogous plots for the charge fraction of the polybase (f_{PB}) under identical conditions. Also, as mentioned in the section titled Methods, due to the charging moves of the polyacid and polybase being independent, the system is not always charge neutral. The net charge in the simulations under the same conditions as this figure is presented in Fig. S3 of the [supplementary material](#).

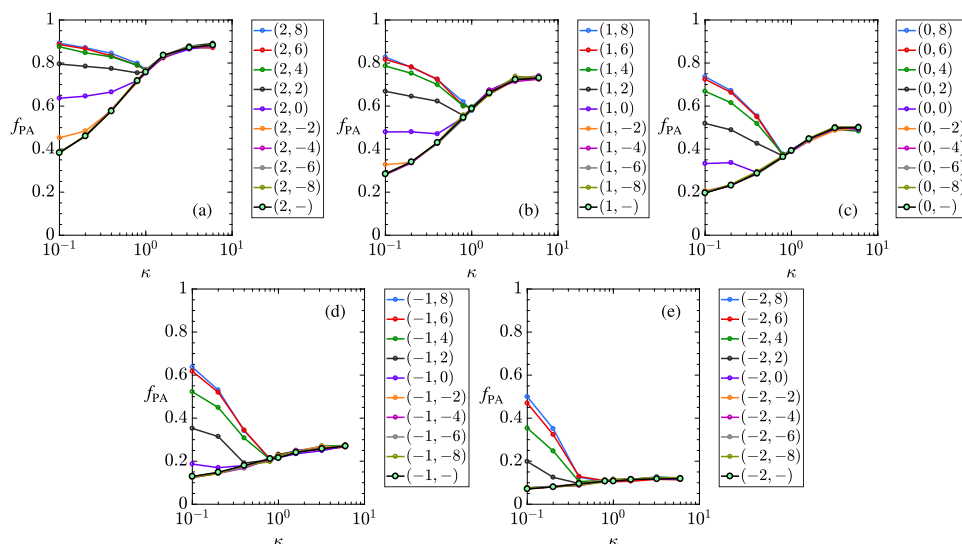


FIG. 4. The equilibrium charge fraction of the polyacid plotted as a function of κ , μ_{PA} , and μ_{PB} . In each panel, the black lines with green circles represent charging behavior of an isolated polyacid, whereas the blue line depicts the maximum extent of associative charging among the set of μ_{PA} and μ_{PB} . The set of μ_{PA} and μ_{PB} is represented in the legend as (μ_{PA}, μ_{PB}) . Note the collapse onto the black line at $\kappa = 1.0$, indicating the absence of associative charging due to interaction with polybase, whereby charging of the polyacid is similar to isolated polyacids and dependent only on the μ_{PA} . Figure S2 of the [supplementary material](#) depicts the charge fraction of polybase seen from this perspective, i.e., variable μ_{PB} and fixed μ_{PA} for each plot (a)–(e).

significant increase in the charge on the polyacid. For example, at $\mu_{PA} = -2$ and $\kappa = 0.1$, it is highly unfavorable for monomers of an isolated polyacid to charge, with the charge fraction with a charge fraction ≈ 0.1 for the polyacid [Fig. 4(e)]. However, the charge fraction of the polyacid increases to ≈ 0.5 when the polyacid interacts with a polybase having a high charging tendency of $\mu_{PB} = 8$ [see Figs. 3(a) and 4(e)]. Similar effects may also be seen for other values of μ_{PA} . For instance, at $\kappa = 0.1$ when the interactions between monomers is relatively long ranged, the charge fraction of the polyacid increases from ≈ 0.2 ($\mu_{PA} = 0$, $\mu_{PB} = -8$), which is the isolated polyacid limit, to ≈ 0.8 ($\mu_{PA} = 0$, $\mu_{PB} = 8$) due to significant charging cooperativity between the polyacid and highly charged polybase [see Figs. 3(a) and 4(c)]. Note that as μ_{PA} increases [from panel (e) to panel (a) in Fig. 4], the extent of charging monotonically increases both for isolated and interacting chains; however, strong electrostatics that would result between negatively charged polyacid monomers and positively charged polybase monomers can lead to a significant enhancement in charging of the polyacid.

As the κ values increase, concomitant with increased salt content and screening, this associative charging of the polyacid due to interaction with the polybase decreases. This behavior is seen in the heat maps of Figs. 3(a)–3(d), where the light green line (corresponding to a half-charged state of the polyacid) is straightened with increases in κ , approaching ideal HH conditions in the absence of inter- and intra-chain association. At salt concentrations above $\kappa = 1$ (representing ~ 0.18 M solution of monovalent salts), the titration curves in Fig. 4 collapse onto the values expected for isolated chains, indicating that at this κ the interaction between polyelectrolytes no longer affects the charging of the polyacid. This absence of association at or near $\kappa = 1$ can also be corroborated with the complexation study performed by Nair *et al.*,⁵⁸ utilizing polyampholytes and strong polyelectrolytes. Such behavior is evident at all μ_{PA} and μ_{PB} (see Fig. 4). Note that this threshold may change significantly if the chains are non-linear, due to modification of the intra-chain interactions. The explicit details involved in the charging of isolated, but connected, chains are a subject of developing research.¹⁹ Hence, at $\kappa = 6$ (Figs. 3

and 4), we see that the charge fraction of polyacid approaches a fixed value for all values of μ_{PB} . There, the charging of the polyacid is governed by μ_{PA} alone, as an increase in the value of μ_{PA} leads to higher charging independent of the polybase charge. Taken as a whole, the results in Figs. 3 and 4 demonstrate a striking result concerning the cooperativity of charging. While individual chains resist charging more than would be expected from the HH equation, associating chains favor charging—often strongly—balancing the intra-chain effects in systems with matching μ and hence enhancing charging for weaker polyelectrolytes through interactions with stronger polyelectrolytes.

Intuition regarding the role of associative interactions in charging weak polyelectrolytes is confirmed by examining the molecular conformations. In Fig. 5, equilibrated configurational snapshots are presented for different (μ_{PA}, μ_{PB}) at $\kappa = 0.1$ on a grid. Light and dark blue represent uncharged and charged monomers of polyacid, while light and dark red represent uncharged and charged monomers of the polybase, respectively. The configurations presented in Fig. 5 represent specific snapshots contributing to the charge fraction data for polyacids presented in Fig. 3(a). An interesting observation that can be taken from the conformations of Fig. 5 is that complexation is seen between polyacids and polybases at (μ_{PA}, μ_{PB}) of (0,8), as well as the symmetric conditions of (8,0); when compared to charge fraction data presented in Fig. 3(a), we would anticipate a polyacid charge fraction polyacid of ~ 0.8 . This suggests that at low screening conditions, complexation between the polyacid and highly charged polybase is induced by promoting charging of the polyacid even when the polyacid might not have a strong tendency to charge on its own. Configurations represented by (0,8), (0,0), and (0,-8) in Fig. 5 show fully complexed, complexing and uncomplexed structures in that order, with the symmetric conditions behaving as expected. Note that only when the μ_{PA} or μ_{PB} value is very low is there an absence cooperative charging and thus no complexation [see, for example, the states $(-8, 8)$ and $(8, -8)$ in Fig. 5].

To further elucidate the complexation behavior, in Figs. 6(a) and 6(b), we present the average distance between

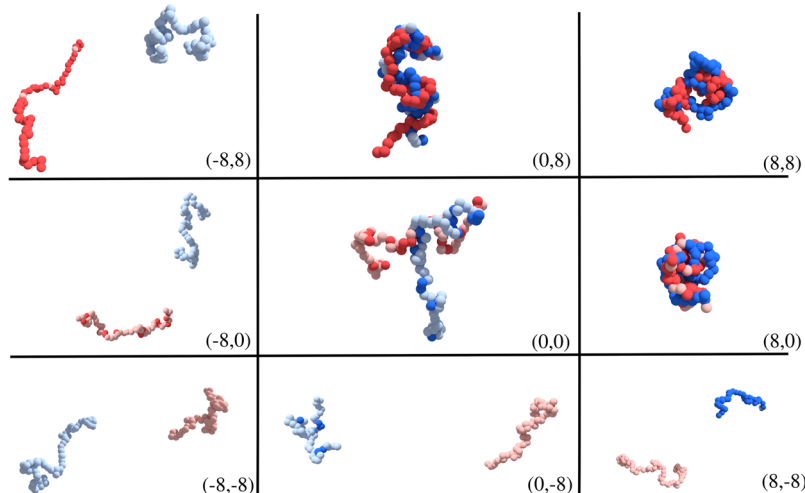


FIG. 5. Equilibrated configurational snapshots of the system for different (μ_{PA}, μ_{PB}) and corresponding $\kappa = 0.1$. Here, the light blue color and light red color represent the uncharged polyacid repeat unit and uncharged polybase repeat unit, respectively, while the dark blue color and dark red color represent the charged polyacid repeat unit and charged polybase repeat unit, respectively. It should be noted that for extreme μ values, for example, $\mu_{PA} = -8$ and $\mu_{PB} = 8$, although the charging tendency for polybase is very high, the very low tendency to charge for polyacid negates any associative charging effects due to polybase even at $\kappa = 0.1$. This is further evidenced by results presented in Fig. 3(a) where negligible charging for a polyacid is observed and thus no complexation is seen between the polyacid and polybase at $(-8,8)$.

the center of mass of the polyacid and polybase, at $\kappa = 0.1$ and $\kappa = 6.0$ respectively. The dark blue region in Fig. 6(a) indicates the formation of a complex, while the red region delimits states where chains do not bind. Note that the center of mass distance at $(\mu_{PA}, \mu_{PB}) = (0,-8)$ and $(0,8)$ in Fig. 6(a) bolsters the evidence of complexation presented in the charging fractions of Fig. 4 and is also in accord with the results presented in Fig. 3(a) which shows associativity in the charging of the polyacid due to complexation with the polybase. However, in the case of $\kappa = 6.0$, the center of mass distance across all values of μ_{PA} and μ_{PB} indicates no complex formation between the polyacid and polybase as represented through the dark red region in Fig. 6(b), and this corroborates the results presented in Fig. 3(d) where no associativity in the charging of the polyacid was indicated.

Information about the structure of polymer complexes may be obtained using the radius of gyration for the polyacid; these data are presented in Figs. 6(c) and 6(d) for $\kappa = 0.1$

and $\kappa = 6.0$, respectively (for the analogous polybase plots, see Fig. S4 of the [supplementary material](#)). An example of the complexation behavior can be noticed in Fig. 6(c) for $\mu_{PA} = 0$ at $\kappa = 0.1$, where at low μ_{PB} values (-8 to -2) the polyacid assumes extended configuration (red region) due to non-associativity with the polybase and because of repulsive interaction between like-charged monomers; however, increasing values of μ_{PB} (2 – 8) corresponds to complex formation (blue region) due to the associative charging of the polyacid as is also indicated in Fig. 3(a). A small island of extended configurations can also be observed in Fig. 6(a), which corresponds to a complex formed by a fully charged polybase with an associatively charged polyacid “coiling” around the extended polybase (see the inset). Conditions where this complex forms are marked in Fig. 6 with an arrow. Note that, due to the symmetric nature of our model, similar complexes also form where the charging potential of the polyacid and polybase is reversed (see Fig. S4 of the [supplementary material](#)).

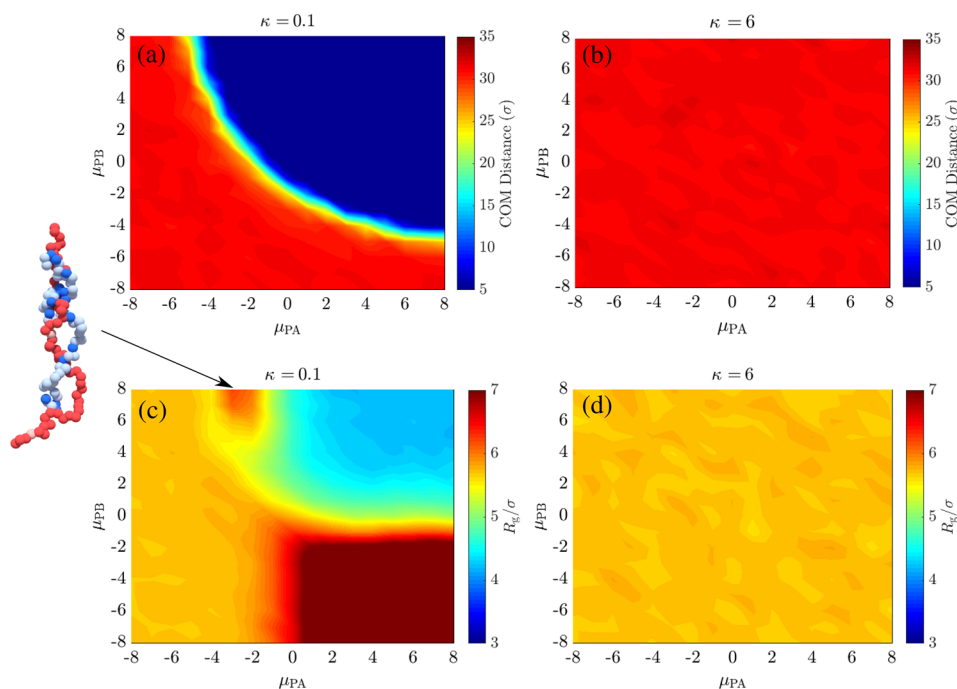


FIG. 6. Panels (a) and (b) depict the distance between centers of mass of polyacids and polybases in the simulation units σ for $\kappa = 0.1$ and $\kappa = 6.0$, respectively. Center of mass distance less than or equal to 5σ is indicative of formation of a complex (collapsed or intermingled structure) between the polyacid and polybase. (c) and (d) show the radius of gyration of the polyacid across a range of μ_{PA} and μ_{PB} for $\kappa = 0.1$ and $\kappa = 6.0$. The complex configuration marked with the arrow depicts the occurrence of a small red colored region in (c) and is a representative of “coiling” of an associatively charged polyacid along the almost fully charged and extended polybase. The radius of gyration of the polybase is calculated under the same conditions as Figs. 6(c) and 6(d) and is presented in Fig. S4 of the [supplementary material](#).

By contrast at $\kappa = 6.0$ presented in Fig. 6(d) no associative interactions are observed, and thus the radius of gyration remains effectively constant across all values of μ_{PA} and μ_{PB} . Note that for $\kappa = 6$ or $\lambda_D = 0.166$, at lower values of μ_{PB} (-8 to -2) and $\mu_{PA} = 0$, the red region corresponding to extension of a polyacid chain (large R_g) is not observed due to extensive screening which results in minimal interaction between the charged monomers of the polyacid. The information in these plots can be corroborated by the comparison of associative conformations (measured by the radius of gyration) for an isolated polyacid at its pK_a in Fig. S5 of the [supplementary material](#). Importantly, collapse of the polyacid in the blue region may only be seen in the presence of a charged polybase.

Thus far we have depicted complexation behavior and associative charging effects through variations in molecular configurations (Figs. 5 and 6) and charge fraction (Figs. 3 and 4). To further elucidate the effects of κ and pH on associative charging, in Fig. 7 we examine the potential energy U as a function of κ at different μ_{PA} and μ_{PB} values. First, it is instructive to focus on the cases where $\mu_{PB} = 8$, as the polybase is nearly fully charged there. We note that there is a decrease in the potential energy approaching lower κ values, which occurs due to the onset of favorable attractive electrostatic interactions. These become more favorable as μ_{PA} increases and the polyacid becomes more likely to charge. Focusing on the opposite case, $\mu_{PB} = -8$, where the polybase is essentially uncharged, shows that increasing the charging tendency of the polyacid (high μ_{PA} value) acts to raise the potential energy, as the intra-chain contributions to energy dominate the system behavior. In between, the system seeks a balance between unfavorable intra-chain charging and favorable associative interactions.

Increases in κ have an interesting nonmonotonic effect, which nonetheless may be understood by thinking about the tug of war between association and charging. Initially, upon increasing κ , all but the lowest-charging polybase conditions exhibit a hill in their energy landscape, representative of increased contributions of repulsive interaction by neighboring similarly charged monomers, while simultaneously the favorable contribution from charge associations is reduced as long-range forces are curtailed. The shift in the peak to right side from panel (a) to panel (e) in Fig. 7 is due to increasing tendency of the polyacid to charge as the μ_{PA} value is increased, which allows associative interactions (which lower the energy) to contribute appreciably even when screened. The cusp is determined by the change from an association-dominated landscape to a repulsive landscape dominated intra-chain interactions, after which increasing κ further decays the energy to its expected background level for two Kremer-Grest-like chains. For paired chains which favor charging, this occurs at $\kappa\sigma = 1$, while it necessarily occurs earlier when one chain is significantly inhibited by low μ . These results are mirrored by the charging landscapes plotted in Fig. 4.

As discussed in the Introduction, the study of polyelectrolyte complexes has been a promising field for last few years with some useful applications emerging in drug delivery,⁵⁹ where pH-responsive complexes could further offer more target-specific tuning for drug delivery.^{24,60,61} Moreover, weak polyelectrolyte complexation could also be utilized in gene delivery, by exploiting the pH drop in the lysosome,^{62–64} for easy release of not only genetic material inside the cell but also drugs and other polypeptides.

Based on our results, we discuss some mechanisms by which that complexes comprised of weak polyelectrolytes

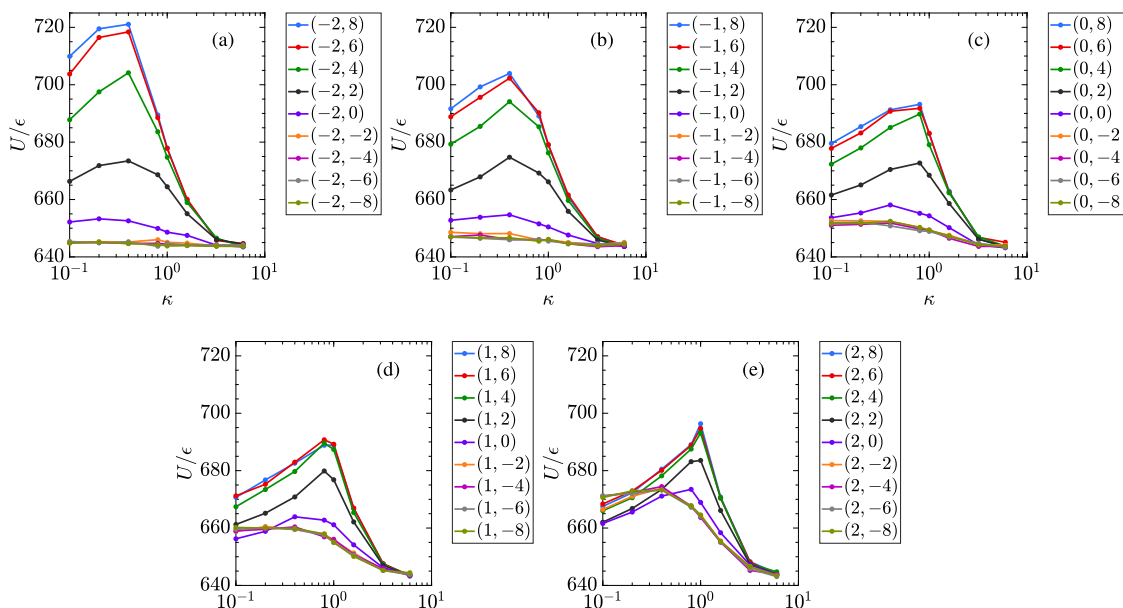


FIG. 7. The equilibrium potential energy of the system, U , plotted as a function of κ , μ_{PA} , and μ_{PB} . The legend in the figure represents (μ_{PA}, μ_{PB}) . Associative charging is favorable due to the decrease in potential energy at smaller κ values. The peak is representative of an increased contribution from the repulsive interactions of monomers (through increased charging) and concomitant reduction of the favorable contribution from charge associations as long-range forces are curtailed. At much larger values $\kappa > 1$, the potential energy values decrease (reduction in intra-chain repulsion) and become independent of κ , μ_{PA} , and μ_{PB} .

may be utilized for drug delivery and controlled release. The results in Figs. 3 and 4 have shown that oppositely charging chains may be tightly wound or uncomplexed structures which suggest drug loading and drug release, respectively, in environmental conditions where (μ_{PA}, μ_{PB}) and κ vary, while the salt concentration in most biologically relevant systems is fixed, and yield Debye lengths larger than the monomer sizes discussed here, the choice of monomer type can be utilized to balance μ_{PA} and μ_{PB} for optimal selective stability. Note that this often requires a good separation in pH between the administered complexes and the therapeutic environment. As suggested by our results, the integrity of a complexed structure may be maintained in many different environments while transporting to the site of drug action if the materials are properly chosen. Considering conditions of $(\mu_{PA}, \mu_{PB}) = (0, 8)$, which are indicative of the limit of a strong polybase complexing with a weak polyacid, in Figs. 3(a) and 6, change in the μ_{PA} value by ± 3 still results in a complexed structure due to associative charging of the polyacid. Using a combination of weak polymers allows this range to be reduced; as seen in Fig. 6(a), the domain of stable complexes is convex, and a window for the stable pH of complexes may be exquisitely tuned by the choice of monomer pK_a and pK_b . This provides additional flexibility in tuning the materials for drug transport to the site of action.

Importantly, for peptide or genetic therapies, one of the polyelectrolytes involved in the complex may be the deliverable itself. Significant recent attention has been given to efficient transfection of the gene during gene delivery.⁶⁵ Polycations have already been used for efficient compaction of DNA through complex formation. However, complexation with a weak polyelectrolyte can also offer further environmental adaptability. Consider, for example, the configurations and charge fraction results presented in Fig. 5 and Figs. 3 and 4, respectively. At high $\mu_{PA} = 8$, the polyacid is in the highly charged polyelectrolyte limit. Now consider two environmental conditions one where it is favorable for a polybase to charge up ($\mu_{PB} = 8$), resulting in a complexed structure with polyacid, and another which provides unfavorable conditions for a polybase to charge ($\mu_{PB} = -8$), resulting in an uncomplexed structure. Hence reversible complexation between a DNA segment (highly charged polyacid) and another degradable, biocompatible, or bespoke weak polybase should be possible, thus readily favoring the release of DNA in one environment despite complexation in another.

Other noteworthy experimental work on pH-responsive complexes has been also conducted in the last few years, e.g., Paloma *et al.*⁶⁶ prepared complexes of chitosan and polyacrylic acid (PAA) to study the release of drug amoxicillin trihydrate and found that the diffusion of amoxicillin trihydrate was controlled only by the swelling of the complexes and that the water uptake was mainly governed by the degree of ionization, and similarly, the experimental study by Win *et al.*⁶⁷ where a polyelectrolyte complex based on phosphorylated chitosan and tripolyphosphate polyanion was developed to study the release of ibuprofen in oral administration and noted that the release rate of ibuprofen at pH 7.4 was higher than pH 1.4, indicating the target specificity or pH responsiveness of the polyelectrolyte complex.

Extensions of our investigations including explicit ions should be able to resolve any limitations on associative charging effects. Though for the monovalent salts the Debye-Hückel potential is able to qualitatively predict experimental behavior,⁶⁸ the assumptions of the Debye-Hückel potential break down fairly quickly for $\kappa > 1$ for multivalent salts as ion correlations would affect the charging behavior in this regime. At very large μ values, approximating the strong polyelectrolyte limit, ion correlations will also become important due to counterion condensation. Such studies would additionally help resolve important questions which remain related to how this mechanism would manifest in experiments on associating polyelectrolytes. Solutions of oppositely charged polymers are often understood through the mechanism of counterion release, proposed in an early simulation study⁸ and expanded in several thermodynamic models for complex coacervation.^{69–72} If weak polyelectrolytes are viewed similarly to strong polyelectrolytes which are solvated, and associate through a combination of strong electrostatic binding and entropic gain from ion release into the surrounding environment, our treatment within this paper presents (in essence) only the energetic contributions as the mechanism of complexation. What is important to note here is that in those models the polymer is assumed to be charged already and can thus bind either to an ion or to the oppositely charged electrolyte, whereas here the polymer is assumed uncharged until it (de)protonates. In this way, there is a fundamental energetic gain that can drive complexation in some conditions. When added salt is present, those ions will undoubtedly provide competitive binding for the polyelectrolytes, but only when charged. It is thus likely that both mechanisms are at play in solutions of oppositely charged weak polyelectrolytes and their gegenions. One theoretical treatment has attempted to explain their mutual influence,⁷³ though molecular simulations should much more clearly elucidate the role of microions in influencing the charging and association of weak polymers. Importantly, we stress that the charging effect and polymer association observed herein should be looked at as an independent effect from that examined in the case of strong polyelectrolyte complexation and that models incorporating weak polyelectrolyte charging effects should also account for the effects of explicit ions. To enable future design and control of polyelectrolyte materials, it will be important to resolve how associative charging, as presented in the current study, fits into the current theoretical picture for polyion association.

CONCLUSIONS

The results of our work serve to demonstrate significant, nontrivial contributions of the local energetic environment to the charging of weak polyelectrolytes. In particular, for weak polyelectrolytes utilized to form complex coacervates, this indicates that sections of the polyelectrolyte may be more highly charged than others, where they come into contact with an oppositely charged chain. Importantly, this can act to stabilize the coacervate, resulting in more strongly associating systems than naïve expectation from strong polyelectrolytes would imply. This presents a testable hypothesis, where strong and weak polyelectrolytes of similar flexibility can be prepared

at a similar linear charge density through modification of polymer sequence or solution conditions and their phase behavior consequently compared.

Importantly, we have shown here that associative interactions can drive the charging of chains with $\mu \leq 0$, which prefer not to charge on their own. This study ignores potentially significant additional effects from the solvation of chains, hydrogen bonding, explicit charge correlations, and topological characteristics. Our work has demonstrated effects that are often ignored in the calculation of, e.g., coacervation properties that can have significant effects on charging and adhesion and thus should be accounted for when designing applications involving weak polyelectrolyte complexes. This has significant implications for the study of polyelectrolyte materials that have been loaded with therapeutics, as in “smart” materials for drug delivery, as the energetic environment for charging has a significant influence on the charge state of the polyelectrolyte complex. Subtle association effects between the polymer(s) and the drug(s) can create substantial shifts in the apparent pK_a of the molecule and polymer, subsequently altering the resulting bioavailability. By the same token, such shifts could be utilized to stabilize polyelectrolyte coacervates, through mutually favorable associative interactions driven by (de)protonation.

SUPPLEMENTARY MATERIAL

See [supplementary material](#) for supporting figures plotting the response of polybase molecules under the conditions outlined in this paper.

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